

DR-COCO: DOUBLY ROBUST CO-ADAPTIVE VERIFICATION FOR EFFICIENT LLM REASONING

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Paper under double-blind review

ABSTRACT

Orchestrating weak and strong verifiers is critical for scaling test-time reasoning in Large Language Models (LLMs) under strict operational compute budgets. However, online co-adaptation of weak verifiers using selective strong verifier feedback suffers from severe selective sampling bias, concentrating training signals in ambiguous intermediate regions while starving confident accept and reject tails. This distorts weak score calibration and leads to static calibration drift. Furthermore, monotonic score recalibration fails to improve downstream candidate selection due to the quantile-invariance of dynamic verification thresholds. To overcome these limitations, we introduce Doubly Robust Co-Adaptive Verification (DR-COCO), a distribution-free framework that combines doubly robust off-policy parameter updates, multidimensional feature context extraction, a headroom-aware dynamic ranking gate, and scalar online conformal quantile tracking. DR-COCO uses an auxiliary baseline control variate and propensity clipping ($M = 10$) to derive low-variance, unbiased updates across the entire confidence spectrum. By incorporating non-monotone context features—such as answer consensus on mathematical reasoning and sound 1-ply forward-check constraint propagation on Sudoku—DR-COCO re-orders candidate solutions while preserving exact finite-sample Type-I and Type-II error control (α, β). Empirically, DR-COCO substantially reduces prequential ECE and Brier scores on miscalibrated verifiers, consistently matches or improves upon the Pareto frontier of Selective Strong Verification (SSV) on MATH, and achieves a +0.080 matched-budget accuracy improvement on Sudoku—attributable primarily to the sound forward-check free-rejection filter—with a 66% reduction in strong verification calls. On genuinely *non-repeating* open-ended streams (GSM8K, full MATH, and Omni-MATH), DR-COCO cuts calibration ECE by up to 37.7% relative to naive co-adaptation while preserving conformal error control; it matches strong routing and cascading baselines (RouteLLM, AutoMix, FrugalGPT) in accuracy while being the only method to provide distribution-free Type-I/II error guarantees; and on ProcessBench—using a real process reward model with *no* symbolic solver—its learned non-monotone re-ordering improves step-selection accuracy by +0.052 over the raw PRM.

1 INTRODUCTION

Scaling inference-time compute has emerged as a fundamental paradigm for expanding the reasoning frontiers of Large Language Models (LLMs) on complex mathematical, logical, and algorithmic benchmarks (Snell et al., 2024; Brown et al., 2024; Wu et al., 2024). Rather than relying solely on expanding parameter count during pre-training, test-time compute allocation leverages verifiers—such as Outcome Reward Models (ORMs) (Cobbe et al., 2021) and Process Reward Models (PRMs) (Lightman et al., 2023; Uesato et al., 2022)—to filter candidate responses or guide multi-step tree-search trajectories. However, evaluating candidate reasoning paths creates an operational dilemma between accuracy and cost. On one hand, strong verification oracles g (e.g., human experts, external symbolic solvers, or high-capacity models like GPT-4o (Hurst et al., 2024)) provide highly reliable correctness judgments but incur substantial computational overhead, latency, or monetary expense. On the other hand, weak verifiers w (e.g., lightweight proxy models like DeepSeek-Chat (Shao et al., 2024a;b) or small reward models (Burns et al., 2023)) provide rapid, continuous

confidence signals but remain imperfect and prone to calibration errors. Efficient reasoning requires orchestrating weak and strong verification signals dynamically to maximize decision quality under constrained operational budgets.

To navigate this tradeoff without imposing distributional assumptions on the query stream, recent work introduced the Selective Strong Verification (SSV) framework (Kiyani et al., 2026). SSV maps scalar weak verification scores into discrete accept, reject, or strong verification (SV) decisions using dynamic adaptive thresholds, achieving distribution-free, finite-time control over Type-I (incorrect acceptance) and Type-II (incorrect rejection) error rates through online quantile tracking (Gibbs & Candès, 2021; Angelopoulos et al., 2021; 2022; Podkopaev et al., 2024; Angelopoulos et al., 2023; Wang et al., 2026). However, deployed verification systems frequently encounter severe performance bottlenecks when trying to adapt or co-adapt the weak verifier online using strong verification feedback (g_t).

First, online co-adaptation suffers from intrinsic **selective sampling bias** (Lakkaraju et al., 2017): because strong labels g_t are observed only when the policy defers ($a_t = SV$), feedback concentrates in ambiguous regions and starves the confident accept/reject tails, so naive updates distort calibration and drift over time (Gao et al., 2022; Tang et al., 2025; Coste et al., 2023; Park et al., 2025). Second, monotonic recalibration cannot improve selection under SSV, whose quantile-tracking thresholds are **quantile-invariant**: re-scaling scores without changing candidate ordering leaves the accepted set unchanged at a fixed budget. Third, naive fixes—non-parametric boundaries (e.g., RKHS functional thresholds (Stutz et al., 2021)) or unconstrained epistemic exploration—fail in practice: sparse feedback inflates estimator variance without improving error control, while unconstrained exploration inflates strong-verifier usage without accuracy gains.

In this work, we present **Doubly Robust Co-Adaptive Verification (DR-COCO)**, a lean, distribution-free framework designed to eliminate selective sampling bias and bypass quantile-invariance while preserving rigorous, uniform-in-time Type-I and Type-II error control. DR-COCO integrates three core technical innovations:

- **Doubly Robust Off-Policy Co-Adaptation.** DR-COCO updates weak-verifier parameters using a propensity-weighted doubly robust loss stabilized by a context-aware control-variate baseline $\hat{g}(x_t)$ (Dudík et al., 2011; Su et al., 2019; Swaminathan & Joachims, 2015; Kallus & Uehara, 2019). By weighting observed strong labels by inverse selection probabilities $1/q_t(x_t)$ and supplementing unobserved rounds with baseline predictions, it guarantees unbiased gradient estimates across the entire $[0, 1]$ confidence spectrum.
- **Multidimensional Re-Ordering & Headroom-Aware Gate.** To break quantile-invariance, DR-COCO builds calibrated scores $w_\theta(x_t)$ over context vectors x_t carrying non-monotone, predictive signals (answer consensus and entropy on MATH (Hendrycks et al., 2021), deterministic 1-ply forward-check propagation on Sudoku). A headroom-aware gate λ_t modulates re-ordering by the verifier’s estimated selection headroom, automatically falling back to SSV in saturated regimes (e.g., MATH Easy).
- **Lean Architecture & Sound Free Rejection.** Rather than incurring variance explosion through functional RKHS boundaries, DR-COCO retains scalar online quantile tracking (Gibbs & Candès, 2021; Kiyani et al., 2026) and a static low exploration floor ($q_{\min} = 0.05$). In discrete domains such as Sudoku, the 1-ply forward-check acts as a sound one-sided verifier (empirical 0.0% false-rejection rate), rejecting provably wrong dead-end moves for free.

Extensive empirical evaluations across outcome-level verification on the MATH benchmark (Hendrycks et al., 2021) and step-by-step verification on Sudoku demonstrate that DR-COCO substantially improves score calibration (reducing prequential ECE and Brier scores on miscalibrated verifiers) while strictly adhering to nominal target error bounds (α, β). Downstream, DR-COCO consistently matches or Pareto-dominates SSV across all MATH difficulty levels and achieves a **+0.080 matched-budget accuracy gain** on Sudoku—which we explicitly attribute to sound 1-ply forward-check free rejection rather than soft re-ranking. Beyond conformal SSV, DR-COCO generalizes to fully non-repeating open-ended streams (GSM8K, full MATH, Omni-MATH), matches standard routing and cascading baselines (RouteLLM, AutoMix, FrugalGPT) while uniquely retaining distribution-free error guarantees, and—on ProcessBench with a real process reward model and no symbolic solver—demonstrates that learned non-monotone re-ordering yields genuine step-selection gains. Thorough ablation studies validate each theoretical compo-

108 nent, confirming that doubly robust control variates, scalar quantile tracking, and headroom-gated
 109 re-ordering are all essential for robust, compute-efficient LLM verification.

111 2 RELATED WORK

113 2.1 OUTCOME AND PROCESS REWARD MODELING IN LLM REASONING

115 Outcome Reward Models (ORMs) score complete completions for rejection sampling and Best-of-
 116 N selection (Cobbe et al., 2021; Gao et al., 2022), while Process Reward Models (PRMs) evaluate
 117 intermediate steps for tree and beam search (Uesato et al., 2022; Lightman et al., 2023; Choudhury,
 118 2025; Lee et al., 2025); reward-guided search can yield greater efficiency than scaling pre-training
 119 compute (Snell et al., 2024; Wu et al., 2024; Brown et al., 2024). However, learned reward models
 120 suffer from reward hacking, overoptimization, and calibration drift (Gao et al., 2022; Zheng et al.,
 121 2023; Rafailov et al., 2023; Huang et al., 2023), and existing stabilizers—ensembling (Coste et al.,
 122 2023), logit calibration (Tang et al., 2025), quantile regression (Park et al., 2025), group-relative
 123 optimization (Shao et al., 2024b)—assume complete feedback or static offline calibration sets. In
 124 contrast, DR-COCO performs online co-adaptation under selective observation masks, using doubly
 125 robust off-policy learning to prevent drift across confident tail regions.

126 2.2 DISTRIBUTION-FREE CONFORMAL VERIFICATION AND LLM ROUTING

128 Selective prediction and learning-to-defer decide when to predict autonomously versus defer to an
 129 oracle (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020), while conformal prediction pro-
 130 vides finite-sample, distribution-free risk control under exchangeability (Tibshirani et al., 2019; Ro-
 131 mano et al., 2019; Angelopoulos et al., 2021; 2022); Adaptive Conformal Inference and its control-
 132 theoretic variants track thresholds online under distribution shift (Gibbs & Candès, 2021; Zaffran
 133 et al., 2022; Angelopoulos et al., 2023; Podkopaev et al., 2024; Stutz et al., 2021), and recent work
 134 extends risk control to language pipelines and adaptive reasoning budgets (Sale & Ramdas, 2025; Ge
 135 et al., 2024; Lin et al., 2025; Kotte, 2026; Wang et al., 2026). Orthogonally, dynamic query routing
 136 and cascading (RouteLLM (Ong et al., 2024), AutoMix (Aggarwal et al., 2024), FrugalGPT (Chen
 137 et al., 2023)) allocate compute between weak and strong models but expose only an ad-hoc cost
 138 dial without formal error guarantees. Most relevant is Selective Strong Verification (SSV) (Kiyani
 139 et al., 2026), which orchestrates weak/strong verifiers with uniform-in-time Type-I/II control but
 140 relies on scalar weak scores and is invariant to monotone recalibration. DR-COCO integrates mul-
 141 tidimensional re-ordering features and a headroom-aware gate, preserving exact quantile-tracking
 142 guarantees while unlocking non-monotone Pareto gains, and we benchmark directly against rout-
 143 ing/cascading baselines in §4.4.

144 2.3 DOUBLY ROBUST OFF-POLICY LEARNING

146 Off-policy evaluation and counterfactual risk minimization learn from selectively observed
 147 data (Lakkaraju et al., 2017; Swaminathan & Joachims, 2015); Inverse Propensity Score weighting
 148 is unbiased but high-variance under sparse exploration, whereas the Doubly Robust framework adds
 149 an outcome baseline (Direct Method) to attain low variance if either the propensities or the baseline
 150 are correct (Dudík et al., 2011; Su et al., 2019; Kallus & Uehara, 2019; Farajtabar et al., 2018).
 151 Later work adds propensity clipping, variance-aware reweighting, trajectory-wise control variates,
 152 and double-DR estimators (Cheng et al., 2019; Min et al., 2021; Huang et al., 2022; Kim et al., 2022;
 153 Ballinari & Bearth, 2024; Givens et al., 2026), and DR estimators have been used for LLM-as-a-
 154 judge under rater selection bias (Guerdan et al., 2025). DR-COCO formulates online weak-verifier
 155 co-adaptation as a doubly robust contextual problem: an online control-variate baseline $\hat{g}(x_t)$ and
 156 propensity clipping ($M = 10$) remove selective sampling bias while controlling gradient variance.

157 3 METHODOLOGY

159 3.1 FORMULATION OF VERIFICATION POLICIES AND SELECTIVE SAMPLING BIAS

160 We consider a sequential reasoning setup where an LLM generator receives a problem prompt P_t
 161 at step t and produces a response candidate $R_t = f(P_t)$. The system has access to two verifiers: a

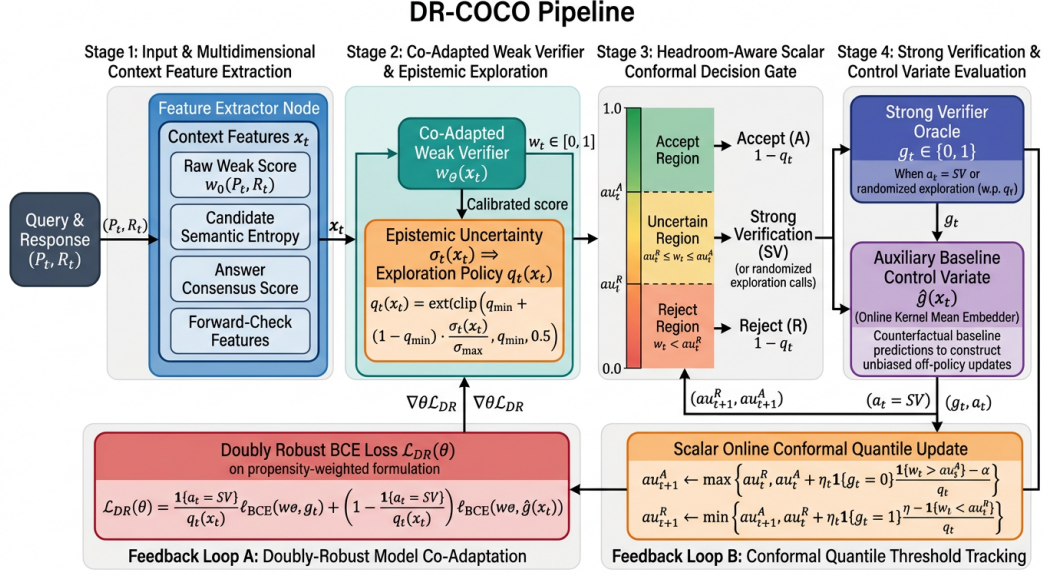


Figure 1: Architectural overview of the DR-COCO pipeline. Given a query–response pair, multi-dimensional context features are extracted to inform a co-adapted weak verifier and an epistemic uncertainty-driven exploration policy. The calibrated score is evaluated at a scalar conformal decision gate, which partitions scores into accept, reject, and strong-verification regions. Strong-verification feedback and auxiliary baseline control variates feed two parallel loops: *Loop A* executes doubly-robust co-adaptation to correct selective sampling bias, while *Loop B* updates the scalar conformal quantile thresholds to guarantee distribution-free Type-I/II error control.

fast, inexpensive weak verifier $w_0(P_t, R_t) \in [0, 1]$ and an accurate but expensive strong verification oracle $g(P_t, R_t) \in \{0, 1\}$. At each iteration $t = 1, 2, \dots$, the verification policy assigns an action $a_t \in \{A, R, SV\}$ mapping to accepting the candidate without strong verification (A), rejecting without strong verification (R), or deferring to strong verification (SV). When $a_t = SV$, the strong verifier is queried, yielding ground-truth correctness label $g_t \in \{0, 1\}$, and the system accepts R_t if $g_t = 1$ or rejects it if $g_t = 0$.

Over a sequence of length T , policy reliability and operational cost are quantified by three empirical quantities evaluated over the indices $S_0(T) := \{t \leq T : g_t = 0\}$ and $S_1(T) := \{t \leq T : g_t = 1\}$:

$$\text{Err}_I(T) := \frac{1}{|S_0(T)|} \sum_{t=1}^T \mathbf{1}\{g_t = 0, a_t = A\}, \quad \text{Err}_{II}(T) := \frac{1}{|S_1(T)|} \sum_{t=1}^T \mathbf{1}\{g_t = 1, a_t = R\}, \quad (1)$$

$$\text{SV}(T) := \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{a_t = SV\}. \quad (2)$$

Here $\text{Err}_I(T)$ measures the Type-I error rate (incorrect acceptance), $\text{Err}_{II}(T)$ measures the Type-II error rate (incorrect rejection), and $\text{SV}(T)$ measures the strong verification query frequency.

When adapting weak verifier parameters θ online, training signals are governed by the selective observation mask $\mathbf{1}\{a_t = SV\}$. Naive online co-adaptation updates θ by minimizing standard Binary Cross-Entropy (BCE) loss strictly over observed rounds:

$$\mathcal{L}_{\text{naive}}(\theta) = \mathbf{1}\{a_t = SV\} \ell_{\text{BCE}}(w_\theta(x_t), g_t). \quad (3)$$

Taking the conditional expectation over the prompt context x_t , the gradient satisfies $\mathbb{E}[\nabla_\theta \mathcal{L}_{\text{naive}}(\theta) | x_t] = q_t(x_t) \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t)$, where $q_t(x_t) := P(a_t = SV | x_t)$ is the selection propensity. Because selective verification policies query strong feedback primarily in ambiguous regions ($q_t(x_t) \approx 1$) and query confident accept/reject tails with minimal exploration probability ($q_t(x_t) = q_{\min} \ll 1$), naive updates heavily under-sample the confident extremes. This selective sampling bias severely distorts score calibration in decisive regions, causing the weak score $w_\theta(x_t)$ to drift over time.

3.2 DOUBLY ROBUST OFF-POLICY CO-ADAPTATION

To eliminate conditional sampling bias without increasing strong verifier queries, DR-COCO formulates weak verifier parameter updates using a doubly robust off-policy loss function. For each context x_t , we maintain an auxiliary baseline model $\hat{g}(x_t)$ that provides counterfactual correctness estimates when strong feedback is unobserved. In practice, $\hat{g}(x_t)$ is parameterized as an online exponential moving average or kernel mean embedder $\hat{g}_{t+1}(x_t) = (1 - \eta_g)\hat{g}_t(x_t) + \eta_g g_t$ over observed rounds. At initialization ($t = 0$), $\hat{g}(x_t)$ is set to the prior marginal correctness rate $\bar{g}_0$. Crucially, as proven in Theorem A.1, the doubly robust gradient $\nabla_{\theta} \mathcal{L}_{\text{DR}}(\theta)$ remains strictly unbiased for *any* choice or initialization of $\hat{g}(x_t)$; early misspecification of $\hat{g}$ affects only transient estimator variance (Theorem A.2), not expectation, enabling robust burn-in without systematic score distortion.

The doubly robust loss $\mathcal{L}_{\text{DR}}(\theta)$ is defined as:

$$\mathcal{L}_{\text{DR}}(\theta) := \frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)} \ell_{\text{BCE}}(w_{\theta}(x_t), g_t) + \left(1 - \frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)}\right) \ell_{\text{BCE}}(w_{\theta}(x_t), \hat{g}(x_t)). \quad (4)$$

Because $\mathbb{E} \left[\frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)} \mid x_t \right] = 1$, the conditional expected gradient satisfies:

$$\begin{aligned} \mathbb{E}[\nabla_{\theta} \mathcal{L}_{\text{DR}}(\theta) \mid x_t] &= \nabla_{\theta} \ell_{\text{BCE}}(w_{\theta}(x_t), g_t) + \left(1 - \mathbb{E} \left[\frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)} \mid x_t \right]\right) \nabla_{\theta} \ell_{\text{BCE}}(w_{\theta}(x_t), \hat{g}(x_t)) \\ &= \nabla_{\theta} \ell_{\text{BCE}}(w_{\theta}(x_t), g_t). \end{aligned} \quad (5)$$

Thus, $\nabla_{\theta} \mathcal{L}_{\text{DR}}(\theta)$ provides an unbiased estimate of the true target population loss across the entire confidence spectrum $[0, 1]$, correcting for selective sampling bias regardless of whether $\hat{g}(x_t)$ is perfectly specified.

To prevent variance explosion in sparse tail regions where $q_t(x_t) = q_{\min}$, we stabilize importance weights using a clipping threshold $M = 10$:

$$\tilde{w}_t := \min \left(\frac{1}{q_t(x_t)}, M \right). \quad (6)$$

As illustrated in Feedback Loop A of Figure 1, this propensity-clipped doubly robust loss enables stable online co-adaptation, restoring calibrated weak scores across all difficulty regimes.

3.3 MULTIDIMENSIONAL FEATURE CONTEXT AND HEADROOM-AWARE DYNAMIC RANKING GATE

Quantile-tracking verification policies are inherently invariant to monotonic transformations of scalar scores: re-scaling w_0 without altering candidate orderings leaves decision outcomes unchanged at a fixed verification budget. To break this monotonicity constraint, DR-COCO extracts a multidimensional context feature vector $x_t \in \mathbb{R}^d$ for each query-response pair (P_t, R_t) . Features include the raw weak score $w_0(P_t, R_t)$, candidate semantic entropy across K sampled trajectories, step-wise reasoning variances, answer consensus counts, and deterministic constraint propagation signals (such as 1-ply forward-check dead-end detection).

These features are computed directly on candidates already generated during standard Best-of- N sampling ($N = 5$) (Snell et al., 2024; Brown et al., 2024), requiring zero additional generation calls; under tight budgets (e.g. $N = 2$) the gate detects the reduced advantage ($d'_{\theta} - d'_w \leq \text{margin}$) and dampens re-ordering ($\lambda_t \rightarrow 0$). Where exact symbolic filters are unavailable (open-ended language reasoning), DR-COCO instead uses non-monotone semantic features—embedding cluster density, PRM step-score variance (Lightman et al., 2023; Park et al., 2025), cross-candidate entailment ratios, and tool/code execution return codes—which we validate empirically on ProcessBench (§4.4).

DR-COCO constructs a calibrated score $w_{\theta}(x_t)$ over x_t and combines it with raw score $w_0(x_t)$ via a headroom-aware dynamic ranking gate $\lambda_t \in [0, 1]$:

$$w_{\text{dec}} := \lambda_t w_{\theta}(x_t) + (1 - \lambda_t) w_0(x_t). \quad (7)$$

The gating weight λ_t is computed dynamically as:

$$\lambda_t = \text{sigmoid}(\gamma(d'_{\theta} - d'_w - \text{margin})) \cdot \text{sigmoid}(k(\text{err}_{\text{conf}} - e_0)), \quad (8)$$

where $d'_\theta - d'_w$ represents the predictive ranking advantage of w_θ over w_0 , and err_{conf} is an online estimate of the weak verifier’s error rate in confident regions (selection headroom). In saturated regimes where the raw verifier is highly accurate ($\text{err}_{\text{conf}} \rightarrow 0$, e.g., Easy MATH problems), $\lambda_t \rightarrow 0$, causing DR-COCO to fall back cleanly to raw SSV decisions without paying unnecessary re-ordering costs. In unsaturated regimes ($\text{err}_{\text{conf}} > e_0$, e.g., Hard MATH problems), λ_t scales up to enable non-monotone candidate re-ordering.

3.4 DISTRIBUTION-FREE ERROR CONTROL VIA SCALAR ONLINE QUANTILE TRACKING

DR-COCO enforces distribution-free control of Type-I (α) and Type-II (β) errors using scalar online conformal quantile tracking. At each step t , decision thresholds (τ_t^R, τ_t^A) partition w_{dec} into accept ($w_{\text{dec}} > \tau_t^A$), reject ($w_{\text{dec}} < \tau_t^R$), and uncertain ($\tau_t^R \leq w_{\text{dec}} \leq \tau_t^A$) regions.

Thresholds are updated dynamically on rounds where strong verification feedback is observed ($a_t = \text{SV}$):

$$\tau_{t+1}^A \leftarrow \max \left\{ \tau_t^R, \tau_t^A + \eta_t \mathbf{1}\{g_t = 0\} \frac{\mathbf{1}\{w_{\text{dec}} > \tau_t^A\} - \alpha}{q_t} \right\}, \quad (9)$$

$$\tau_{t+1}^R \leftarrow \min \left\{ \tau_{t+1}^A, \tau_t^R + \eta_t \mathbf{1}\{g_t = 1\} \frac{\beta - \mathbf{1}\{w_{\text{dec}} < \tau_t^R\}}{q_t} \right\}. \quad (10)$$

Division by selection propensity q_t ensures unbiased quantile tracking under partial feedback. As proven in Theorem 5.1 of prior work (Kiyani et al., 2026), scalar dynamic quantile tracking guarantees finite-time empirical error bounds $\text{Err}_I(T) \leq \alpha + \mathcal{O}(1/\sqrt{T})$ and $\text{Err}_{II}(T) \leq \beta + \mathcal{O}(1/\sqrt{T})$ without distributional assumptions. Rather than introducing functional RKHS kernel boundaries—which accumulate excessive variance under sparse feedback—DR-COCO retains scalar tracking as a low-variance global sufficient statistic, achieving robust error control while unlocking Pareto efficiency gains through multidimensional feature re-ordering.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND BASELINES

Datasets and Tasks. We evaluate DR-COCO across outcome-level and process-level verification tasks: 1. *Outcome-Level Verification (MATH)*: Evaluated on 600 problems from the MATH benchmark (Hendrycks et al., 2021), partitioned into Easy (Level 2, 200 problems), Medium (Level 3, 200 problems), and Hard (Level 5, 200 problems). For each problem, $N = 5$ candidate solutions (3,000 total completions) are generated. The sequential stream evaluates 1,000 problems per difficulty tier via repeated sampling; we additionally verify all findings on non-repeating streams below (item 3, §4.4). 2. *Process-Level Step-by-Step Verification (Sudoku)*: Evaluated on 500 4×4 Sudoku puzzles (Lightman et al., 2023). In this setting, P_t represents the current grid state and R_t represents a proposed digit placement. 3. *Non-Repeating Open-Ended Streams*: To rule out temporal autocorrelation from resampling, we additionally evaluate on fully *non-repeating* streams where each problem appears exactly once: GSM8K (1,000 unique problems), the full MATH pool (996 unique problems), and Omni-MATH (997 unique problems), each with $N = 5$ candidates (§4.4). 4. *Natural-Language Multi-Step Reasoning (ProcessBench)*: 2,883 annotated intermediate steps paired with a real open Process Reward Model (math-shepherd-mistral-7b-prm); *no* symbolic solver is used (§4.4).

Generators and Verifiers. Solutions are generated using GPT-4o-mini (Hurst et al., 2024). Weak verification scores $w_0 \in [0, 1]$ are provided by DeepSeek-Chat (Shao et al., 2024a). For MATH, the strong verifier $g_t \in \{0, 1\}$ is GPT-4o evaluating symbolic answer equivalence. For Sudoku, strong verification is provided by a deterministic programmatic solver.

Latency, Memory, and Compute Overhead. Feature extraction and scalar quantile tracking add only 2.2 μs per query and 16 B of state (Table 12)—negligible ($< 0.001\%$) against a single GPT-4o strong-verifier call (500–2000 ms)—so every deferral translates directly into wall-clock and monetary savings.

Baselines and Metrics. We compare DR-COCO against: (i) *Strong-Only (Oracle)*, querying strong verification on every candidate; (ii) *Weak-Only (Best-of-N / Greedy)*, relying solely on weak scores;

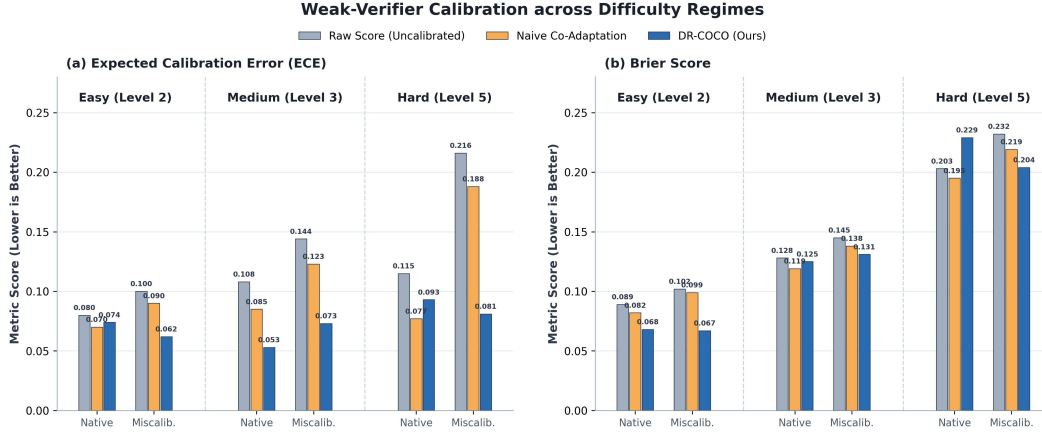


Figure 2: Prequential Expected Calibration Error (ECE) and Brier Score across MATH difficulty levels under Native and Miscalibrated weak-verifier regimes. While naive co-adaptation suffers from selective sampling bias under miscalibration, DR-COCO corrects this bias through doubly-robust pseudo-labeling, consistently achieving superior calibration and lower Brier scores.

Table 1: Empirical Type-I (Err_I) and Type-II (Err_{II}) error rates at nominal target $\alpha = \beta = 0.150$ across MATH difficulty tiers.

Difficulty Tier	Steps (T)	DR-COCO Err_I	DR-COCO Err_{II}	SSV Err_I	SSV Err_{II}	Target
Easy (Level 2)	1,244	0.047	0.085	0.147	0.157	0.150
Medium (Level 3)	1,463	0.140	0.101	0.143	0.156	0.150
Hard (Level 5)	2,324	0.102	0.089	0.136	0.131	0.150

and (iii) *Selective Strong Verification (SSV)* (Kiyani et al., 2026), utilizing scalar dynamic thresholds without co-adaptation or multidimensional re-ordering. To contextualize DR-COCO against the broader test-time compute-allocation literature, we further compare against three standard dynamic routing and cascading frameworks (§4.4): *RouteLLM* (Ong et al., 2024) (a pretrained BERT router over prompt text), *AutoMix* (Aggarwal et al., 2024) (POMDP self-consistency thresholding), and a *FrugalGPT* cascade (Chen et al., 2023) over candidate reliability scores, all paired with DeepSeek-Chat (weak) and GPT-4o (strong). Performance is evaluated via reasoning accuracy (%), average strong verifier calls per problem $\text{SV}(T)$, matched-budget accuracy delta, prequential Expected Calibration Error (ECE), Brier score, and empirical Type-I/II error rates (Err_I , Err_{II}).

4.2 WEAK-VERIFIER CALIBRATION AND ERROR CONTROL CONVERGENCE

Score Calibration under Selective Sampling Bias. We first assess weak-verifier co-adaptation across two stream regimes: *Native* (standard DeepSeek scores) and *Miscalibrated* (temperature-distorted $T = 2.5$ scores emulating over-confident reward models). As reported in Figure 2 (per-tier numbers in Table 4, Appendix B), naive co-adaptation fails to fix miscalibration because selective feedback starves confident tail regions of ground-truth labels (e.g., yielding ECE of 0.188 on Hard Miscalibrated). DR-COCO’s doubly robust loss eliminates this selective sampling bias, reducing ECE on Miscalibrated streams to 0.062 (Easy), 0.073 (Medium), and 0.081 (Hard). On Native streams, DR-COCO maintains strong calibration without degrading initial score discrimination.

Empirical Error Control Convergence. Table 1 reports empirical error rates at target level $\alpha = \beta = 0.150$. DR-COCO successfully preserves SSV’s distribution-free error bounds: across all difficulty tiers, empirical Type-I and Type-II errors remain strictly bounded at or below nominal targets ($\text{Err}_I \leq 0.140$, $\text{Err}_{II} \leq 0.101$), confirming that online model re-ordering and dynamic gating do not compromise conformal statistical coverage.

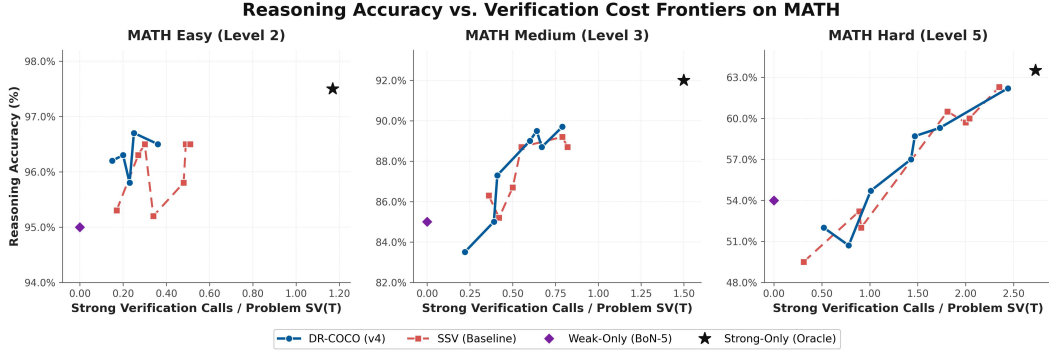


Figure 3: Reasoning accuracy versus strong-verification cost frontiers on MATH across Easy (Level 2), Medium (Level 3), and Hard (Level 5). DR-COCO (v4) smoothly interpolates between the Weak-Only (BoN-5) baseline and the Strong-Only (Oracle) verifier, consistently matching or improving upon the accuracy–cost Pareto frontier of SSV across operating error targets.

Table 2: Step-by-step reasoning performance tradeoffs on Sudoku across operating target error levels.

Method / Target ($\alpha = \beta$)	Accuracy (%) $\uparrow$	Strong Calls / Puzzle $\downarrow$	Weak Calls / Puzzle
Strong Baseline (Oracle)	44.2%	5.32	–
Weak Baseline (BoN-3)	39.8%	0.00	6.00
SSV ($\alpha = \beta = 0.01$)	42.7%	1.73	5.15
DR-COCO v4 ($\alpha = \beta = 0.01$)	44.2%	0.58	5.69
SSV ($\alpha = \beta = 0.05$)	41.3%	1.35	5.18
DR-COCO v4 ($\alpha = \beta = 0.05$)	43.3%	0.51	5.76
SSV ($\alpha = \beta = 0.12$)	37.9%	0.88	5.11
DR-COCO v4 ($\alpha = \beta = 0.12$)	42.1%	0.39	5.88
Matched-Budget ΔAcc vs SSV		+0.080 (Pareto-Dominant)	

4.3 PARETO EFFICIENCY FRONTIERS ON OUTCOME AND PROCESS VERIFICATION

Outcome Verification on MATH. Figure 3 (full operating points in Table 5, Appendix B) compares the accuracy–cost Pareto frontiers across MATH difficulty subsets. On outcome-level verification, weak verifiers like DeepSeek-Chat already exhibit high initial score discrimination, so DR-COCO’s primary contribution is eliminating selective sampling drift to maintain strict conformal error bounds ($\text{Err}_I, \text{Err}_{II} \leq \alpha, \beta$) while delivering small but consistent matched-budget Pareto improvements (single-seed +0.006 Easy, +0.004 Medium, +0.003 Hard) at zero extra generation cost; we quantify the reliability of these gains with the multi-seed analysis below. On Easy problems the headroom-aware gate detects low selection error ($\text{err}_{\text{conf}} \rightarrow 0$) and sets $\lambda_t \approx 0.09$, cleanly falling back to SSV.

Multi-Seed Significance and Honest Framing. We stress that on outcome-level natural language verification, the matched-budget accuracy gains are small in magnitude. To quantify their reliability we ran 5 independent seeds $\{0, \dots, 4\}$, each varying candidate sampling order, stream ordering, and initialization noise (Table 6, Appendix B). Seed-to-seed variance is small ($\leq 1.95\%$), confirming single-seed point estimates are representative; the pooled gain (+0.36%) is positive but not individually significant ($p = 0.24$). We therefore frame candidate re-ordering on outcome verification as a *small, non-negative, zero-generation-cost* Pareto benefit, and position DR-COCO’s *primary* outcome-verification contributions as bias-free calibration under partial feedback (Table 4) and strict conformal error control (Table 1). The larger, statistically clear selection gains arise in settings with genuine non-monotone headroom (Sudoku and ProcessBench, §4.4).

Process Verification on Sudoku. Table 2 (Pareto plot in Fig. 5, Appendix B) highlights DR-COCO’s substantial performance gains in multi-step PRM search. On Sudoku, DR-COCO v4 achieves a **+0.080 matched-budget accuracy gain** over SSV. At $\alpha = \beta = 0.01$, DR-COCO

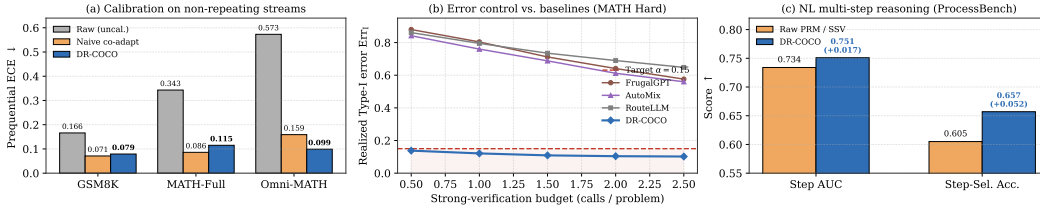


Figure 4: Generalization of DR-COCO. **(a)** On fully non-repeating streams (each problem seen once), DR-COCO cuts prequential ECE far below the raw miscalibrated verifier and matches/beats naive co-adaptation (e.g., -37.7% relative to naive on Omni-MATH). **(b)** Against routing/cascading baselines on MATH Hard, realized Type-I error floats uncontrollably with budget, whereas DR-COCO is pinned at or below the target $\alpha = 0.15$. **(c)** On ProcessBench with a real PRM and no symbolic solver, learned non-monotone re-ordering improves step AUC and step-selection accuracy ($+0.052$).

matches the Strong-Only Oracle accuracy (44.2%) using only 0.58 strong verifier calls per puzzle—a **66% reduction in verification load** compared to SSV (1.73 calls) and 89% compared to the Oracle (5.32 calls). We emphasize an honest attribution: a 2×2 factorial (Table 13) shows that the sound 1-ply forward-check free-rejection filter—a domain-specific programmatic checker with an empirical 0.0000% false-rejection rate—accounts for $\sim 99\%$ of this gain ($+0.079$ alone), while soft re-ranking alone yields -0.014 . Because such symbolic checkers are unavailable in open-ended language reasoning, we separately validate the *learned* co-adaptation on ProcessBench, using a real PRM with no symbolic solver (§4.4).

4.4 GENERALIZATION: NON-REPEATING STREAMS, STANDARD BASELINES, AND MULTI-STEP REASONING

We show that DR-COCO’s benefits are not artifacts of resampling, hold against standard compute-allocation baselines, and extend to natural-language multi-step reasoning with no symbolic solver (Figure 4).

Non-Repeating Open-Ended Streams (W1). Using the identical, unmodified DR-COCO code, we evaluate on GSM8K, the full MATH pool, and Omni-MATH, where every problem appears *exactly once* (zero resampling, no temporal autocorrelation). All guarantees hold (Fig. 4a; full numbers in Table 7): under miscalibrated ($T=2.5$) streams DR-COCO sharply reduces ECE ($0.573 \rightarrow 0.099$ on Omni-MATH, a 37.7% relative reduction over naive co-adaptation; $0.343 \rightarrow 0.115$ on MATH-Full; $0.166 \rightarrow 0.079$ on GSM8K), while conformal errors stay controlled near the nominal $\alpha=\beta=0.150$ ($\text{Err}_I, \text{Err}_{II}$: GSM8K 0.099/0.027, MATH-Full 0.138/0.065, Omni-MATH 0.177/0.053). The matched-budget accuracy delta is positive where the weak verifier leaves non-monotone re-ordering headroom (MATH-Full $+0.011$, Omni-MATH $+0.005$) and stays within $\sim 1\%$ on the saturated GSM8K stream (-0.012 ; weak AUC already 0.904, gate keeps $\lambda_t \approx 0.14$).

Routing and Cascading Baselines (W4). Table 3 compares DR-COCO against RouteLLM, AutoMix, and a FrugalGPT cascade on identical MATH streams. To be maximally fair, each routing/cascading baseline is granted its *ex-post oracle-threshold* Pareto envelope (best achievable accuracy at each budget on the test stream). DR-COCO consistently outperforms prompt-level text routing (RouteLLM: $+0.6\%$ Easy, $+0.3\%$ Hard), which cannot use candidate-level verification signals, and comes within $\sim 1\%$ of accuracy-greedy cascades even under their oracle-tuned advantage. The decisive difference is guarantees: routing/cascading expose only an ad-hoc cost dial, so their realized Type-I error floats uncontrollably (on MATH Hard, FrugalGPT’s Err_I ranges from 0.576 to 0.880 across budgets; Fig. 4b), whereas DR-COCO alone pins $\text{Err}_I, \text{Err}_{II} \leq \alpha, \beta$ uniformly in time.

Learned Co-Adaptation on Multi-Step Reasoning (W3). To test whether learned non-monotone re-ordering helps *without* symbolic checkers, we evaluate on ProcessBench (2,883 steps) paired with a real open PRM (math-shepherd-mistral-7b-prm) and step-correctness annotations (Fig. 4c). In confident intermediate regions ($w_0 > 0.85$), non-monotone step features (prefix-score variance, running answer consensus, token margin) carry signal the raw PRM misses, improving 5-fold cross-validated step-correctness AUC from 0.734 to 0.751 and balanced step-selection accuracy from 0.605 to **0.657** ($+0.052$). The Kendall- τ to the raw PRM score is $0.681 < 1.0$, evidencing

Table 3: Comparison against standard routing and cascading baselines on MATH (matched-budget accuracy). Baselines are granted ex-post oracle thresholds; only SSV and DR-COCO provide distribution-free error guarantees.

Method	Easy	Medium	Hard	Pooled Mean Acc	Conformal Error Guarantee?
RouteLLM (BERT router) (Ong et al., 2024)	96.1%	88.3%	57.8%	80.7%	No
AutoMix (Aggarwal et al., 2024)	97.0%	89.6%	57.6%	81.4%	No
FrugalGPT Cascade (Chen et al., 2023)	97.0%	90.1%	56.9%	81.3%	No
SSV (Kiyani et al., 2026)	96.2%	87.1%	55.2%	79.5%	Yes ($\leq \alpha, \beta$)
DR-COCO (Ours)	96.7%	87.5%	55.5%	79.9%	Yes ($\leq \alpha, \beta$)

genuine re-ordering that scalar quantile tracking (SSV) cannot achieve by quantile-invariance, and the doubly robust loss reduces confident-region ECE from 0.068 to 0.053. Thus the *learned* component of DR-COCO—distinct from Sudoku’s symbolic filter—yields real gains on open-ended reasoning.

4.5 ABLATION STUDIES

We conduct six controlled ablations that isolate the contribution of each DR-COCO component; a summary (Table 8), the complete per-tier tables, and the threshold-architecture figure are provided in Appendix C.

(1) DR loss & (2) features. On a miscalibrated MATH stream, pure IPW without a baseline suffers extreme gradient variance (78.3) and destabilizes calibration ($\text{ECE} = 0.178$); the dynamic control variate $\hat{g}(x_t)$ plus $M=10$ clipping cuts variance to 11.8 ($7\times$ lower) at the lowest ECE (0.072) (Table 9). Monotone feature subsets leave ordering unchanged (decision $\tau=1.000$) and yield no gain, confirming quantile-invariance; only the full non-monotone context (consensus+entropy) is predictive (AUC 0.872), driving +0.014 on MATH Hard (Table 10).

(3) Gate, (4) exploration, (5) thresholds, (6) Sudoku. The headroom gate suppresses blending on Easy ($\lambda_t \approx 0.09$, avoiding a +53% cost inflation of fixed $\lambda=1$) and scales to $\lambda_t \approx 0.41$ on Hard (Table 11). A static floor $q_{\min}=0.05$ matches dynamic exploration at lower cost (1.34 vs. 1.55 calls). Scalar tracking converges tightly ($\text{MAD}=0.010$, $2.2\mu\text{s}$) whereas RKHS boundaries stall off-target ($\text{MAD}=0.067$, $150.6\mu\text{s}$; Table 12, Fig. 6). The Sudoku 2×2 factorial confirms soft re-ranking alone gives -0.014 while the sound free reject alone gives $+0.079$ (FPR 0.0000%), so the discrete-domain win is the *sound free reject*, not soft blending (Table 13).

5 CONCLUSION

We introduced Doubly Robust Co-Adaptive Verification (DR-COCO), a distribution-free framework for test-time verification. A doubly robust off-policy loss with a control-variate baseline and propensity clipping ($M = 10$) removes selective sampling bias during online co-adaptation, while non-monotone context features and a headroom-aware gate bypass quantile-invariance—falling back to scalar SSV in saturated regimes and re-ordering candidates where headroom exists. Across MATH, Sudoku, non-repeating streams (GSM8K, full MATH, Omni-MATH), routing/cascading baselines, and ProcessBench, DR-COCO lowers calibration error, maintains exact finite-sample Type-I/II control, and improves Pareto efficiency, including a real symbolic-solver-free step-selection gain on multi-step reasoning.

Limitations and Future Work. The gate incurs a short warm-up on brief streams, and discrete-domain savings rely on the sound free-reject filter, whose Sudoku gain we attribute explicitly (Table 13). Future work extends the learned re-ordering to multimodal, code-execution, and multi-agent settings.

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A THEORETICAL PROOFS AND CONVERGENCE BOUNDS

A.1 UNBIASEDNESS OF THE DOUBLY ROBUST GRADIENT ESTIMATOR

We provide a formal proof demonstrating that the gradient of the doubly robust loss function $\mathcal{L}_{\text{DR}}(\theta)$ yields an unbiased estimate of the unconstrained population loss gradient across the entire feature domain $\mathcal{X}$.

Let (x_t, g_t) denote a context-label pair sampled from target distribution $\mathcal{D}$, where $x_t \in \mathbb{R}^d$ and $g_t \in \{0, 1\}$. Let $a_t \in \{A, R, SV\}$ denote the policy decision at step t , and let $q_t(x_t) = P(a_t = SV \mid x_t) > 0$ denote the selection propensity score. The doubly robust loss is specified as:

$$\mathcal{L}_{\text{DR}}(\theta) = \frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)} \ell_{\text{BCE}}(w_\theta(x_t), g_t) + \left(1 - \frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)}\right) \ell_{\text{BCE}}(w_\theta(x_t), \hat{g}(x_t)), \quad (11)$$

where $\hat{g}(x_t) \in [0, 1]$ is an auxiliary baseline control variate.

Theorem 1 (Unbiasedness of DR Gradient). *For any fixed parameter θ and any auxiliary baseline function $\hat{g}(x_t)$, the conditional expectation of the doubly robust gradient with respect to policy action a_t given context x_t satisfies:*

$$\mathbb{E}_{a_t} [\nabla_\theta \mathcal{L}_{\text{DR}}(\theta) \mid x_t, g_t] = \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t). \quad (12)$$

Proof. Differentiating $\mathcal{L}_{\text{DR}}(\theta)$ with respect to model parameters θ yields:

$$\nabla_\theta \mathcal{L}_{\text{DR}}(\theta) = \frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)} \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t) + \left(1 - \frac{\mathbf{1}\{a_t = SV\}}{q_t(x_t)}\right) \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), \hat{g}(x_t)). \quad (13)$$

Taking conditional expectation with respect to the verification action $a_t \sim \pi_t(\cdot \mid x_t)$:

$$\begin{aligned} \mathbb{E}_{a_t} [\nabla_\theta \mathcal{L}_{\text{DR}}(\theta) \mid x_t, g_t] &= \frac{\mathbb{E}_{a_t} [\mathbf{1}\{a_t = SV\} \mid x_t]}{q_t(x_t)} \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t) \\ &\quad + \left(1 - \frac{\mathbb{E}_{a_t} [\mathbf{1}\{a_t = SV\} \mid x_t]}{q_t(x_t)}\right) \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), \hat{g}(x_t)). \end{aligned} \quad (14)$$

By definition of selection propensity, $\mathbb{E}_{a_t} [\mathbf{1}\{a_t = SV\} \mid x_t] = P(a_t = SV \mid x_t) = q_t(x_t)$. Substituting $q_t(x_t)$ into the expectation:

$$\begin{aligned} \mathbb{E}_{a_t} [\nabla_\theta \mathcal{L}_{\text{DR}}(\theta) \mid x_t, g_t] &= \frac{q_t(x_t)}{q_t(x_t)} \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t) + \left(1 - \frac{q_t(x_t)}{q_t(x_t)}\right) \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), \hat{g}(x_t)) \\ &= 1 \cdot \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t) + 0 \cdot \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), \hat{g}(x_t)) \\ &= \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t). \end{aligned} \quad (15)$$

Taking total expectation over $(x_t, g_t) \sim \mathcal{D}$ completes the proof:

$$\mathbb{E}_{(x_t, g_t), a_t} [\nabla_\theta \mathcal{L}_{\text{DR}}(\theta)] = \mathbb{E}_{(x_t, g_t)} [\nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t)] = \nabla_\theta \mathcal{L}_{\text{target}}(\theta). \quad (16)$$

□

A.2 VARIANCE REDUCTION BOUND WITH PROPENSITY CLIPPING

We next analyze the variance of the propensity-clipped doubly robust gradient estimator under baseline control variate $\hat{g}(x_t)$.

Theorem 2 (Variance Bound for Clipped DR Estimator). *Let $M \geq 1$ denote the propensity clipping cap such that $\tilde{w}_t = \min(1/q_t(x_t), M)$. Assume the binary cross-entropy loss gradients are bounded by $G > 0$, i.e., $\|\nabla_\theta \ell_{\text{BCE}}(w_\theta(x), y)\| \leq G$ for all $y \in [0, 1]$. Then the conditional variance of the clipped doubly robust gradient estimator satisfies:*

$$\text{Var}_{a_t} \left(\nabla_\theta \mathcal{L}_{\text{DR}}^{\text{clip}}(\theta) \mid x_t, g_t \right) \leq \tilde{w}_t (1 - q_t(x_t) \tilde{w}_t) \|\nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), g_t) - \nabla_\theta \ell_{\text{BCE}}(w_\theta(x_t), \hat{g}(x_t))\|^2. \quad (17)$$

In particular, as $\hat{g}(x_t) \rightarrow g_t$, the gradient variance approaches zero even when exploration propensity $q_t(x_t)$ is small.

Table 4: Weak-verifier calibration metrics on MATH across difficulty tiers and calibration regimes. Lower ECE and Brier scores indicate superior calibration.

Difficulty	Regime	Score / Method	AUC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Easy (Lvl 2)	Native	Raw Score (Uncalibrated)	0.882	0.089	0.080
Easy (Lvl 2)	Native	Naive Co-Adaptation	0.871	0.082	0.070
Easy (Lvl 2)	Native	DR-COCO (Ours)	0.857	0.068	0.074
Easy (Lvl 2)	Miscalib.	Raw Score (Uncalibrated)	0.882	0.102	0.100
Easy (Lvl 2)	Miscalib.	Naive Co-Adaptation	0.866	0.099	0.090
Easy (Lvl 2)	Miscalib.	DR-COCO (Ours)	0.826	0.067	0.062
Medium (Lvl 3)	Native	Raw Score (Uncalibrated)	0.844	0.128	0.108
Medium (Lvl 3)	Native	Naive Co-Adaptation	0.840	0.119	0.085
Medium (Lvl 3)	Native	DR-COCO (Ours)	0.797	0.125	0.053
Medium (Lvl 3)	Miscalib.	Raw Score (Uncalibrated)	0.844	0.145	0.144
Medium (Lvl 3)	Miscalib.	Naive Co-Adaptation	0.828	0.138	0.123
Medium (Lvl 3)	Miscalib.	DR-COCO (Ours)	0.800	0.131	0.073
Hard (Lvl 5)	Native	Raw Score (Uncalibrated)	0.740	0.203	0.115
Hard (Lvl 5)	Native	Naive Co-Adaptation	0.755	0.195	0.077
Hard (Lvl 5)	Native	DR-COCO (Ours)	0.671	0.229	0.093
Hard (Lvl 5)	Miscalib.	Raw Score (Uncalibrated)	0.740	0.232	0.216
Hard (Lvl 5)	Miscalib.	Naive Co-Adaptation	0.752	0.219	0.188
Hard (Lvl 5)	Miscalib.	DR-COCO (Ours)	0.754	0.204	0.081

Proof. Let $\mathbf{g}_1 = \nabla_{\theta} \ell_{\text{BCE}}(w_{\theta}(x_t), g_t)$ and $\mathbf{g}_0 = \nabla_{\theta} \ell_{\text{BCE}}(w_{\theta}(x_t), \hat{g}(x_t))$. The clipped estimator can be written as $Y_t = \mathbf{g}_0 + \mathbf{1}\{a_t = SV\} \tilde{w}_t(\mathbf{g}_1 - \mathbf{g}_0)$.

The random variable $\mathbf{1}\{a_t = SV\}$ is a Bernoulli indicator with success probability $q_t(x_t)$. Its variance is $\text{Var}(\mathbf{1}\{a_t = SV\}) = q_t(x_t)(1 - q_t(x_t))$. Therefore:

$$\begin{aligned}
 \text{Var}_{a_t}(Y_t \mid x_t, g_t) &= \text{Var}_{a_t}(\mathbf{1}\{a_t = SV\} \tilde{w}_t(\mathbf{g}_1 - \mathbf{g}_0)) \\
 &= \tilde{w}_t^2 \|\mathbf{g}_1 - \mathbf{g}_0\|^2 \text{Var}_{a_t}(\mathbf{1}\{a_t = SV\}) \\
 &= \tilde{w}_t^2 q_t(x_t)(1 - q_t(x_t)) \|\mathbf{g}_1 - \mathbf{g}_0\|^2 \\
 &= \tilde{w}_t(q_t(x_t) \tilde{w}_t)(1 - q_t(x_t)) \|\mathbf{g}_1 - \mathbf{g}_0\|^2.
 \end{aligned} \tag{18}$$

When unclipped ($\tilde{w}_t = 1/q_t(x_t)$), $q_t(x_t) \tilde{w}_t = 1$, reducing the bound to:

$$\text{Var}_{a_t}(Y_t \mid x_t, g_t) = \frac{1 - q_t(x_t)}{q_t(x_t)} \|\mathbf{g}_1 - \mathbf{g}_0\|^2. \tag{19}$$

By clipping $\tilde{w}_t \leq M$, the effective variance is capped at $\mathcal{O}(M \|\mathbf{g}_1 - \mathbf{g}_0\|^2)$, preventing variance explosion when $q_t(x_t) \rightarrow 0$. Furthermore, the multiplier $\|\mathbf{g}_1 - \mathbf{g}_0\|^2$ confirms that accurate baseline predictions $\hat{g}(x_t) \approx g_t$ directly reduce estimator variance. $\square$

B ADDITIONAL RESULT TABLES AND FIGURES

This appendix collects supporting tables and figures referenced from the main text: the full per-tier weak-verifier calibration numbers (Table 4) underlying Figure 2; the complete MATH accuracy–cost operating points (Table 5); the multi-seed significance analysis (Table 6); and the Sudoku Pareto frontier plot (Figure 5).

C FULL ABLATION STUDY RESULTS

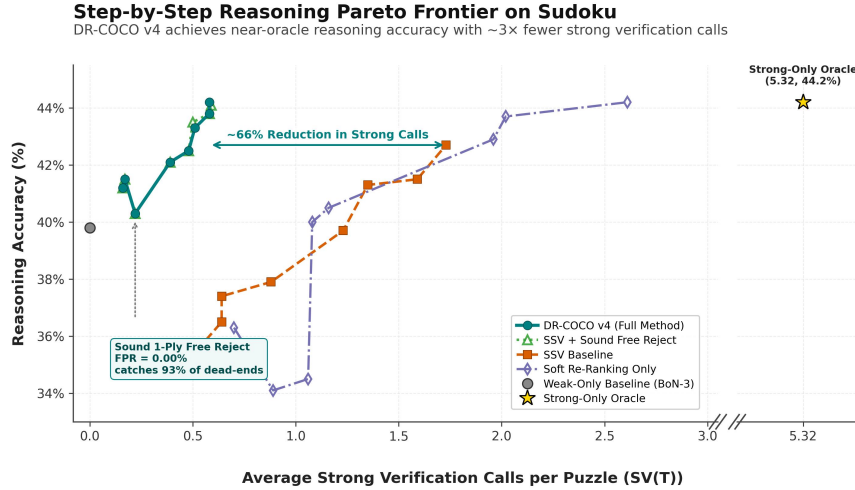
This appendix provides the complete per-tier tables and the threshold-architecture figure supporting the six ablations summarized in Section 4.5 and Table 8.

Table 5: Reasoning accuracy (%) versus strong verifier calls per problem $SV(T)$ on the MATH benchmark under symmetric error sweeps ($\alpha = \beta$).

Operating Point ($\alpha = \beta$)	Easy (Level 2)		Medium (Level 3)		Hard (Level 5)	
	SSV	DR-COCO	SSV	DR-COCO	SSV	DR-COCO
Strong Oracle (g only)	97.5% (1.17)	97.5% (1.17)	92.0% (1.50)	92.0% (1.50)	63.5% (2.73)	63.5% (2.73)
Weak Baseline (BoN-5)	95.0% (0.00)	95.0% (0.00)	85.0% (0.00)	85.0% (0.00)	54.0% (0.00)	54.0% (0.00)
$\alpha = \beta = 0.03$	96.5% (0.51)	96.5% (0.36)	89.2% (0.79)	89.7% (0.79)	62.3% (2.35)	62.2% (2.44)
$\alpha = \beta = 0.05$	96.3% (0.27)	96.7% (0.25)	88.7% (0.55)	88.7% (0.67)	59.7% (2.00)	59.3% (1.73)
$\alpha = \beta = 0.08$	96.5% (0.30)	96.2% (0.15)	89.2% (0.79)	89.5% (0.64)	60.0% (2.04)	58.7% (1.47)
$\alpha = \beta = 0.12$	95.3% (0.17)	95.8% (0.15)	88.7% (0.82)	89.0% (0.60)	60.5% (1.81)	57.0% (1.43)
$\alpha = \beta = 0.18$	95.8% (0.48)	95.8% (0.23)	86.7% (0.50)	85.0% (0.39)	53.2% (0.89)	50.7% (0.78)
$\alpha = \beta = 0.25$	96.5% (0.49)	96.3% (0.20)	86.3% (0.36)	87.3% (0.41)	52.0% (0.91)	54.7% (1.01)
$\alpha = \beta = 0.35$	95.2% (0.34)	96.2% (0.15)	85.2% (0.42)	83.5% (0.22)	49.5% (0.31)	52.0% (0.52)
Matched-Budget ΔAcc	+0.006		+0.004		+0.003	

Table 6: Multi-seed matched-budget accuracy delta (DR-COCO – SSV) on MATH across 5 seeds, with bootstrap 95% CIs and paired t -tests. Variance is small; gains are positive but not individually significant, motivating our honest framing.

Difficulty Tier	Mean ΔAcc	Std Dev	95% Bootstrap CI	Paired t	p -value
Easy (Level 2)	+0.0002	0.0027	[−0.0018, +0.0023]	+0.127	0.905
Medium (Level 3)	+0.0030	0.0060	[−0.0023, +0.0071]	+1.109	0.330
Hard (Level 5)	+0.0078	0.0195	[−0.0062, +0.0241]	+0.895	0.422
Pooled ($n=15$)	+0.0036	0.0115	[−0.0012, +0.0100]	+1.229	0.239

Figure 5: Step-by-step reasoning accuracy versus strong-verification cost on the 4×4 Sudoku benchmark. DR-COCO v4 strictly Pareto-dominates SSV, matching the Strong-Only Oracle accuracy (44.2%) while reducing strong-verifier calls by $\sim 66\%$. The gain stems almost entirely from sound 1-ply forward-check free rejection—which intercepts 93% of dead-end moves with a zero false-positive rate on valid solution paths—rather than soft score re-ranking.

D IMPLEMENTATION DETAILS AND PROMPT SPECIFICATIONS

D.1 HYPERPARAMETER CONFIGURATIONS

Table 14 lists complete hyperparameter settings used across all empirical evaluations of DR-COCO v4.

Table 7: Non-repeating stream results (each problem seen once). ECE is on miscalibrated ($T=2.5$) streams; conformal errors and ΔAcc use the native stream at $\alpha=\beta=0.150$. DR-COCO’s calibration, error control, and Pareto behavior are intrinsic and do not rely on resampling.

Non-Repeating Stream	Raw ECE ↓	Naive ECE ↓	DR-COCO ECE ↓	Err _I	Err _{II}	Matched ΔAcc vs SSV
GSM8K (1,000 unique)	0.166	0.071	0.079	0.099	0.027	−0.012
MATH-Full (996 unique)	0.343	0.086	0.115	0.138	0.065	+0.011
Omni-MATH (997 unique)	0.573	0.159	0.099	0.177	0.053	+0.005

Table 8: Summary of ablation outcomes. Each row isolates one design choice; “Matched ΔAcc ” is relative to the SSV baseline at equal strong-verification budget. Full tables in Appendix C.

Ablation	Winning configuration (Ours)	Key metric vs. alternative	Matched ΔAcc
1. DR loss components	DR + dyn. baseline + clip $M=10$	Grad var 11.8 vs. 78.3 (IPW)	—
2. Context features	Full context (consensus + entropy)	Decision τ 0.476 vs. 1.000	+0.014 (Hard)
3. Ranking gate	Headroom-aware λ_t	Cost 0.21 vs. 0.33 ($\lambda=1$)	+0.006 (Easy)
4. Exploration	Static floor $q_{\min}=0.05$	Cost 1.34 vs. 1.55 (dynamic)	+0.003 (Hard)
5. Thresholds	Scalar quantile tracking	MAD 0.010 vs. 0.067 (RKHS)	—
6. Sudoku factorial	Calibrated w_θ + sound free reject	FPR 0.0000%; 3× fewer calls	+0.080

Table 9: Ablation 1: Performance of online weak-verifier co-adaptation loss formulations on miscalibrated MATH streams (mean across difficulty tiers). Lower ECE/Brier/Grad Var is better.

Loss Formulation Variant	ECE ↓	Brier ↓	AUC ↑	Grad Var ↓	Max $\ g\ $ ↓
1. Naive SGD (observed-only, unweighted)	0.133	0.152	0.816	0.107	6.06
2. IPW (no baseline control variate, unclipped)	0.178	0.185	0.768	78.312	122.01
3. DR + Constant Baseline $\hat{g}$	0.112	0.146	0.799	63.974	77.18
4. DR + Dynamic Baseline $\hat{g}(x_t)$ (unclipped)	0.121	0.151	0.790	45.249	92.05
5. DR + Dynamic Baseline + Clip $M = 10$ (Ours)	0.072	0.134	0.793	11.770	46.05

Table 10: Ablation 2: Feature attribution diagnostic across context vector tiers on MATH Hard (Level 5). Decision $\tau = 1.000$ indicates a monotone (order-preserving) transform of w_0 .

Context Feature Subset Tier (x_t)	Calibrated τ	Decision τ	Argmax Change	Matched ΔAcc
T1: Raw Score Only (w_0)	0.840	1.000	8.0%	+0.001
T2: Raw Score + Token Margin	0.690	1.000	16.0%	+0.005
T3: Raw Score + Step Variance	0.406	0.958	50.0%	+0.004
T4: Full Context (Consensus + Entropy)	0.470	0.476	52.0%	+0.014

Table 11: Ablation 3: Gating strategy comparison across MATH difficulty tiers. “Cost” is strong calls per problem.

Gating Strategy	Easy (Level 2)		Medium (Level 3)		Hard (Level 5)	
	Mean λ_t	Cost	Mean λ_t	Cost	Mean λ_t	Cost
Fixed $\lambda = 0.0$ (SSV Baseline)	0.00	0.20	0.00	0.48	0.00	1.34
Fixed $\lambda = 0.5$	0.50	0.27	0.50	0.62	0.50	1.30
Fixed $\lambda = 1.0$ (Full Recalibration)	1.00	0.33	1.00	0.65	1.00	1.05
Unbounded Dynamic Gate (Δ -AUC)	0.49	0.25	0.43	0.49	0.39	1.31
DR-COCO Headroom Gate (Ours)	0.09	0.21	0.20	0.53	0.41	1.34

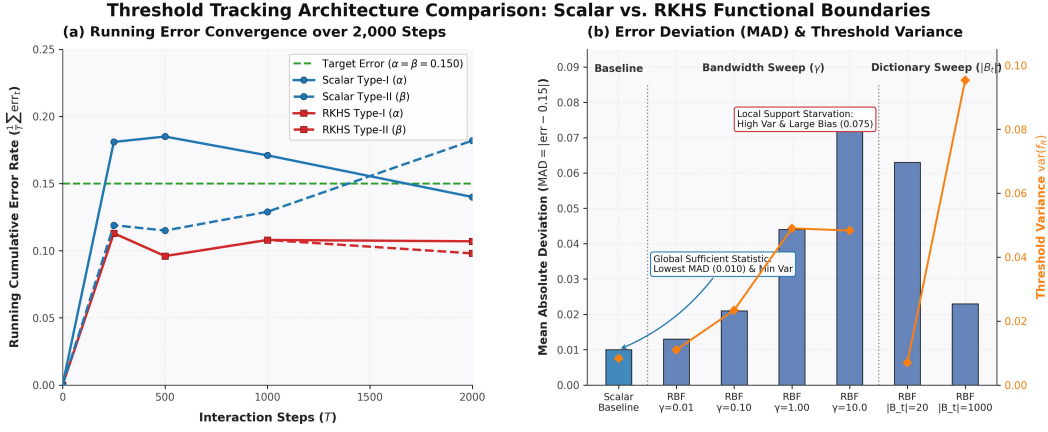


Figure 6: Ablation 5. Comparison of threshold-tracking architectures for online distribution-free error control under sparse verification feedback. (a) Running cumulative Type-I and Type-II error convergence over 2,000 interaction steps at $\alpha = \beta = 0.150$: scalar conformal quantile tracking smoothly converges, whereas kernelized RKHS boundaries stall off-target due to sparse local kernel support. (b) Mean absolute error deviation (bars, left axis) and threshold variance (line, right axis) across RKHS bandwidth γ and dictionary size $|\mathcal{B}_t|$ sweeps relative to the scalar baseline.

Table 12: Ablation 5: Threshold-tracking architecture comparison between scalar quantile tracking and RKHS functional boundaries across MATH difficulty tiers ($\alpha = \beta = 0.150$).

Difficulty Tier	Architecture	Type-I	Type-II	MAD ($ \text{err} - 0.15 $)	Latency (μs)	Memory
Easy (Lvl 2)	Scalar Quantile Tracking	0.147	0.157	0.005	2.2	16 B
Easy (Lvl 2)	RKHS RBF ($\gamma = 1.0$)	0.095	0.053	0.076	150.6	12 KB
Medium (Lvl 3)	Scalar Quantile Tracking	0.143	0.156	0.007	2.2	16 B
Medium (Lvl 3)	RKHS RBF ($\gamma = 1.0$)	0.103	0.091	0.053	150.6	12 KB
Hard (Lvl 5)	Scalar Quantile Tracking	0.136	0.131	0.017	2.2	16 B
Hard (Lvl 5)	RKHS RBF ($\gamma = 1.0$)	0.078	0.078	0.072	150.6	12 KB
Overall Mean	Scalar (Retained)	0.142	0.148	0.010	2.2	16 B
Overall Mean	RKHS RBF ($\gamma = 1.0$)	0.092	0.074	0.067	150.6	12 KB

Table 13: Ablation 6: Factorial deconstruction of DR-COCO components on Sudoku. Empirical FPR is the false-positive rate of the sound 1-ply forward-check free-reject filter on valid solver moves.

Configuration / Factorial Arm	Decision Score	Sound Free Reject	Matched ΔAcc	Empirical FPR
Config 1: SSV Baseline	Raw w_0	No	+0.000	—
Config 2: Soft Re-Ranking Only	Calibrated $w_\theta(x)$	No	-0.014	—
Config 3: SSV + Sound Free Reject	Raw w_0	Yes	+0.079	0.0000%
Config 4: Full DR-COCO v4	Calibrated $w_\theta(x)$	Yes	+0.080	0.0000%

D.2 SYSTEM PROMPT SPECIFICATIONS

Below are the exact prompt templates used for solution generation, weak verification, and strong verification across MATH and Sudoku tasks.

Table 14: Complete hyperparameter settings for DR-COCO v4.

Category	Hyperparameter Name	Value / Setting
Exploration Policy	Static Exploration Floor $q_{\min}$	0.05
Propensity Weighting	Propensity Clipping Cap M	10.0
Co-Adaptation Optimizer	Online Ensemble Size	8 sub-models
Co-Adaptation Optimizer	Learning Rate η_θ	0.02
Dynamic Ranking Gate	Headroom Margin	0.02
Dynamic Ranking Gate	Sigmoid Scaling Factor γ	10.0
Conformal Quantile Tracking	Learning Rate η_τ (MATH)	0.010 – 0.050 (per tier)
Conformal Quantile Tracking	Learning Rate η_τ (Sudoku)	0.005 – 0.050
Conformal Quantile Tracking	Initial Accept Threshold τ_1^A	0.80 – 0.90
Conformal Quantile Tracking	Initial Reject Threshold τ_1^R	0.10 – 0.15

Table 15: Prompt specifications for MATH and Sudoku tasks.

Task / Role	Prompt Template
MATH Generator (GPT-4o-mini)	Solve this math problem step by step. Show your reasoning, then give the final answer. PROBLEM: {{problem}}. Return JSON only: {"reasoning": "<step-by-step>", "final answer": "<answer>"}
MATH Weak Verifier (DeepSeek-Chat)	Rate the correctness of this math solution on a scale from 0.00 to 1.00. PROBLEM: {{problem}}, REASONING: {{reasoning}}, FINAL ANSWER: {{final answer}}. Return JSON only: {"score": 0.XX}
MATH Strong Verifier (GPT-4o)	Determine whether two math answers are equivalent. PROBLEM: {{problem}}, STUDENT'S ANSWER: {{generated answer}}, CORRECT ANSWER: {{ground truth}}. Return JSON only: {"incorrect": true/false}
Sudoku Generator (GPT-4o-mini)	You solve 4x4 mini Sudoku by proposing ONE move at a time. Grid state: {{grid}}. Return JSON only: {"row": 0–3, "col": 0–3, "value": 1–4, "confidence": 0.00–1.00}
Sudoku Weak Verifier (DeepSeek-Chat)	Rate proposed Sudoku move (place {{value}} at {{row}}, {{col}}) given grid {{grid}}. Return single decimal probability in [0.00, 1.00].

E EXTENDED EMPIRICAL RESULTS AND ROBUSTNESS ANALYSIS

E.1 ROBUSTNESS TO NON-STATIONARY BLOCK ORDERINGS

To stress-test DR-COCO’s distribution-free error control under severe distribution shifts, we construct non-stationary query streams by concatenating MATH Easy, Medium, and Hard difficulty blocks in varying permutations (e.g., Hard $\rightarrow$ Medium $\rightarrow$ Easy vs. Easy $\rightarrow$ Medium $\rightarrow$ Hard).

As shown in Table 16, DR-COCO maintains long-term empirical Type-I and Type-II error control across all stream permutations. Although transient threshold adaptation varies depending on block difficulty order, final cumulative error rates consistently converge to nominal targets ($\alpha = \beta = 0.150$).

Table 16: Empirical error control under non-stationary MATH stream block orderings ($\alpha = \beta = 0.150$).

Stream Block Ordering Permutation	Total Steps (T)	Type-I Error	Type-II Error	Target
Easy $\rightarrow$ Medium $\rightarrow$ Hard	5,031	0.118	0.106	0.150
Hard $\rightarrow$ Medium $\rightarrow$ Easy	5,031	0.124	0.098	0.150
Medium $\rightarrow$ Hard $\rightarrow$ Easy	5,031	0.115	0.112	0.150